

NIHAL ERNEST

Gilroy, CA | U.S. Citizen | +1 (714) 854-2722 | nihal.ernest@gmail.com | [Portfolio](#) | [LinkedIn](#)

EDUCATION

University of California, Santa Cruz (UCSC)

- **M.S. Computer Science** GPA: 3.85 Sep. 2024 – Mar. 2026
- **B.S. (Honors) Computer Science** GPA: 3.62 Jul. 2020 – Jun. 2024

EXPERIENCE

Stealth Startup

Jun. 2025 – Jun. 2026

Full Stack/AI Engineer

- Architected a full-stack platform automating real estate investment workflows, with a **Python/Flask** REST API, **SQLite** data layer, and an LLM agent (**OpenClaw** + **Claude**) exposed via Slack through **MCP** on **AWS**. Directly informed **multi-million dollar real estate decisions**.
- Leveraged **SQLite** caching and parallel processing across data ingestion pipelines to reduce an 8-hour process to **under 20 minutes (~24x speedup)**.
- Designed **RESTful** API routes aggregating multi-source geospatial and third-party real estate data, with authentication, observability, and a **git-versioned prompt manager** supporting controlled LLM iteration.

Legal Forensics AI

Jun. 2024 – Sep. 2024

Software Engineering Intern

- Owned end-to-end development of a full-stack web app enabling legal researchers to query **millions of case documents** in natural language, with an **Angular/TypeScript** frontend and **Python** backend powered by **Llama 3** on **AWS Bedrock**.
- Replaced hours of manual keyword filtering with **LLM-driven semantic search** by integrating the Court Listener **REST API**, **surfacing relevant precedents in seconds**.

Xen.AI

Jun. 2023 – Jun. 2024

Software Engineering Intern

- Built multi-tenant **Angular/TypeScript** frontends and **Python** backend function apps on **Microsoft Azure** for **enterprise pilots** in banking and healthcare.
- Integrated **OpenAI GPT-4** and **SQL** databases via **REST** endpoints, implementing **Azure** authentication and search services with **tenant-isolated access controls** to meet banking compliance requirements.

RESEARCH

Sloth Lab, UC Santa Cruz

Sep. 2025 – Jul. 2026

Graduate Researcher

- Built a **natural-language generation pipeline** for **Selma**, a multi-agent simulation of opinion dynamics and group discussion, converting structured interaction events into spoken agent dialogue while maintaining **context across conversation turns**.
- **Co-authored a research paper** on the system, accepted for presentation at the **AIIDE 2026** conference.

PROJECTS

Fishing Platform

Feb. 2026 – Present

- Built a full-stack catch-logging platform (**React** + **Python** REST API), **actively used** by anglers to log catch data into a database-backed metadata layer for historical tracking and cross-user comparison.
- Developed a **recommendation engine** that analyzes historical catch data to suggest fishing plans by time, location, gear, and target species, iterated through ongoing **user feedback**.

Formula Forecast

Sep. 2025 – Present

- Architected a full-stack F1 prediction app (**React** + **FastAPI** + **PyTorch**) with a custom neural network (128→64→32) and a custom Huber-based loss function, achieving **89% accuracy within 3 positions**.
- Built an end-to-end data pipeline ingesting race telemetry via the **FastF1 API**, engineering 9 features per driver-race, and serving predictions through **REST** endpoints, using **time-aware cross-validation** to prevent data leakage.

TECHNICAL SKILLS

Languages Python, JavaScript, TypeScript, Java, C/C++ , HTML/CSS, SQL (PostgreSQL, MySQL, SQLite)

Frameworks React, Angular, React Native, FastAPI, Flask, PyTorch

Cloud & Tools AWS, Azure, Git, GitHub Actions (CI/CD), Linux, REST APIs, OpenAI, Llama, Supabase, Selenium